Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | sahaj-bhadja | sahaj-bhadja | sahaj-bhadja

Summary

Software engineer with 2+ years building internal web platforms, TypeScript applications, SQL-backed services, and workflow automation using React, Next.js, Python, REST APIs, Git, cloud infrastructure, and AI-assisted development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant Azure SQL and Blob Storage services for 500+ users, optimizing schemas, indexed queries, and access patterns to reduce internal dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI workflows processing 5,000+ monthly jobs with explicit states, validation, reconciliation, monitoring, and failure isolation, achieving 99%+ reliability while reducing recurring administrative effort.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask APIs handling 10,000+ daily requests, translating instructor requirements into intuitive upload, review, history, feedback, and reporting workflows for 150+ active users.
- Automated concurrent processing for 1,000+ uploaded files using multithreading, secure ingestion, AWS RDS, and fault-tolerant execution, reducing turnaround by 60% while consistently isolating operational production failures.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform services and 20+ automated PyTest and Mockito suites through Jenkins CI/CD, improving nightly release consistency across Windows, Linux, and macOS with detailed actionable failure reports.
- Optimized REST API and ETL workflows using object-oriented patterns, AWS Glue, S3, and SQL, improving response times by 30%, pipeline efficiency by 68%, and data-processing systems maintainability.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a modular MERN application with REST APIs, WebSockets, and eight services supporting 2M+ synchronized records while translating defined requirements into responsive customer and inventory workflows reliably.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving integration consistency, defect resolution, and production delivery reliability for engineering teams.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs for a booking application supporting scheduling, payments, and status updates across 50,000+ records with synchronized Firebase-backed workflows, validation controls, and clear operational state tracking.
- Implemented webhook-based payment synchronization and root-cause troubleshooting, reducing manual handling by 70% while improving transaction visibility, error recovery, and responsiveness across distributed application components during production operations.

Skills

Web Development: TypeScript, JavaScript, React, Next.js, Node.js, HTML5, CSS3, responsive UI, state management, REST APIs, WebSockets, JSON
Data & Automation: SQL, PostgreSQL, MySQL, Azure SQL, SQL Server, MongoDB, Firebase, data validation, query optimization, workflow automation, reporting, ETL
AI & Development Tools: Claude, GPT-4o, LLM APIs, prompt design, output review, Cursor, VS Code, Git, GitHub, code reviews, documentation
Backend & Delivery: Python, Java, Flask, FastAPI, microservices, AWS, Azure, Docker, Jenkins, GitHub Actions, CI/CD, debugging, root-cause analysis, Agile/Scrum

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven, Node.js*
- Built an AI-assisted database workspace supporting 10+ systems, accelerating SQL authoring, schema exploration, validation, and data management through a unified TypeScript and Java interface for internal users.

HMIS Live | *TypeScript, MERN, REST APIs, Appwrite, Twilio*
- Built operational scheduling, record-access, and notification workflows supporting 100+ daily appointments, reducing administrative workload by 30% through responsive dashboards, automated reminders, and monitored API integrations across teams.

FinNexus Live | *Next.js, TypeScript, MongoDB, Plaid, WebSockets*
- Built a scalable operations platform supporting 50,000+ users and 1M+ transactions, combining secure APIs, real-time synchronization, advanced filtering, and automated deployment workflows reliably across cloud environments.